Covid-19 Global Impact Dashboard



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OWID_COVID_DATASET (COVID GLOBAL IMPACT ANALYSIS)

1. **Why was it necessary to filter out OWID_* ISO codes, trim the final 14 days, and reindex the dataset to a continuous daily calendar before calculating rolling averages and lagged metrics?**

Answer:

In the raw Our World in Data dataset (429,435 rows across 67 columns), 34,124 rows (7.95%) are non-country regional, continental, or income-tier rollups prefixed with OWID_ (such as OWID_WRL, OWID_EUR, and OWID_HIC). Filtering these out via ~covid["iso_code"].str.startswith("OWID_", na=False) isolated the 395,311 sovereign country rows and prevented multi-counting before subsetting the data to the seven target countries (BRA, DEU, GBR, IND, ISR, USA, and ZAF), which initially yielded 11,726 rows. From there, trimming the final TAIL = 14 days (cutting the dataset at 2024-07-31 from the raw maximum date of 2024-08-14) removed incomplete provisional reporting at the end of the timeline. Most importantly, reindexing the subset onto a rectangular MultiIndex grid (pd.MultiIndex.from_product([ISO, full])) guaranteed exactly 1,668 continuous calendar days per country (11,676 total rows). Without this calendar-safe reindexing, any missing reporting dates would cause Pandas .rolling(7) and .shift(21) operations to align by row index rather than true calendar days, distorting both the 7-day rolling averages and the 21-day infection-to-death lag

2. **How does the 21-day lagged Case Fatality Rate (cfr_lagged_28d) differ mathematically and epidemiologically from the naive Case Fatality Rate (cfr_naive_28d), and why does this distinction matter on the dashboard?**

Answer:

A naive Case Fatality Rate (cfr_naive_28d) divides the 28-day rolling sum of deaths (d28) by the 28-day rolling sum of cases from the exact same time window (c28_now). Epidemiologically, this creates a severe distortion during the onset and decline of pandemic waves because COVID-19 mortalities clinically lag behind initial case diagnoses by roughly three weeks. When a new wave surges, dividing current deaths (which reflect infections from weeks ago) by rapidly exploding current cases artificially depresses the naive CFR; conversely, as a wave subsides,

current deaths remain elevated while new cases drop, causing the naive CFR to spike falsely. By engineering cfr_lagged_28d—which divides the current 28-day rolling death sum (d28) by the 28-day rolling case sum shifted backward by LAG = 21 days (c28_now.groupby(df["iso_code"]).shift(21))—the pipeline aligns fatalities with the infection cohort that actually produced them. As shown on Page 3 of the Power BI dashboard (*"Aligning deaths to infections ~21 days earlier changes the fatality story"*), plotting cfr_lagged_28d alongside cfr_naive_28d across countries like Brazil, Germany, India, and Israel reveals the true trajectory of viral lethality over time.

3. How did you engineer the 14-day case doubling time (doubling_time_14d) and automated wave peak detection (is_wave_peak) in Python to handle noisy real-world time-series data?

Answer:

Real-world COVID-19 case reporting suffers from administrative batching and flat or declining post-wave periods where simple day-over-day growth ratios either divide by zero or produce negative doubling times. To solve this, doubling_time_14d was derived by taking the natural logarithm of the smoothed daily cases per million (np.log(np.clip(s.values, 1e-6, None))) and fitting a linear regression slope (r = np.polyfit(x[i - win : i], y[i - win : i], 1)[0]) over a 14-day sliding window (DBL_WIN = 14).

The doubling time was then calculated as $\ln(2)/r$ strictly when the daily exponential growth rate r exceeded 10^{-4}, automatically assigning NaN during flat or declining phases so dashboard aggregations would not be skewed by negative or infinite values. Similarly, to identify distinct infection waves objectively across countries without manual hardcoding, the flag_peaks function evaluated new_cases_smoothed_per_million_7d over a centered 43-day rolling window (2 * k + 1 where PEAK_K = 21 days) and flagged a date as a wave peak (is_wave_peak = 1) only if it was the local maximum within ± 21 days and met a minimum severity threshold of PEAK_FLOOR_PM = 50.0 cases per million.

This rule cleanly detected 12 wave peaks in the United Kingdom, 11 each in Germany and Israel, 9 each in Brazil and the United States, 6 in South Africa, and 3 in India.

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4. What did your focused hypothesis analysis reveal about the relationship between vaccination rollout speed (days_10_to_50) and post-50% peak mortality (peak_deaths_pm_post50), and how do demographic confounders explain the outliers

Answer:

To test the core portfolio question—*"Did faster vaccination blunt later-wave deaths?"*—rollout velocity was defined as the number of calendar days required for a country to progress from 10% to 50% fully vaccinated (days_10_to_50), which was plotted against the highest 7-day rolling deaths per million recorded after crossing the 50% threshold (peak_deaths_pm_post50). South Africa (ZAF) was transparently excluded from the scatter plot (n = 6 of 7 countries plotted) because it never crossed the 50% full-vaccination threshold within the dataset window.

Among the remaining six nations, comparing the United Kingdom (87 days from 10% to 50%, 3.46 peak deaths per million) against the United States (127 days, 7.80 peak deaths per million) supports the hypothesis that faster rollouts reduced later-wave mortality.

However, the quadrant analysis highlights why vaccination speed alone represents an association rather than direct causation. Specifically, Germany (76 days, 5.25 peak deaths per million) and Israel (63 days, 6.89 peak deaths per million) landed in the *"FAST + HIGH (outlier — why?)"* quadrant due to key confounders: Germany had the oldest population in the cohort (median_age = 46.6 years), while Israel experienced an extreme Omicron infection surge exceeding 7,700 daily cases per million in early 2022. Conversely, India (156 days, 0.79 peak deaths per million) and Brazil (148 days, 4.52 peak deaths per million) fell into the slower-rollout quadrants, where India's much younger demographic profile (median_age = 28.2 years) and under-ascertainment of deaths (0.00% excess mortality data coverage in OWID, compared to 12.53% in DEU, GBR, ISR, USA, and ZAF) acted as major confounding factors.

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5. How did you structure the data exports on Day 4 (dash_fact.csv, dash_country.csv, and dash_tidy_long.csv) to support a multi-page interactive Power BI dashboard without introducing aggregation errors?

Answer:

A common pitfall in business intelligence dashboards is feeding a single flat time-series table into both daily trend charts and country-level summary scatter plots, which forces BI tools to sum or average pre-calculated ratios across thousands of rows.

To prevent this, the Day 4 export pipeline separated the curated outputs from Day 2 into three purpose-built tables with strict schema validation (assert sizes.nunique() == 1 and assert not fact.duplicated(["iso_code", "date"]).any()). First, dash_fact.csv (11,676 rows × 11 columns) preserved the wide daily time series grain—containing new_cases_per_million_7d, new_deaths_per_million_7d, people_fully_vaccinated_per_hundred_7d, cfr_lagged_28d, cfr_naive_28d, doubling_time_14d, and is_wave_peak—to power the multi-line wave and fatality comparisons on Pages 2 and 3. Second, dash_country.csv (7 rows × 10 columns) isolated the one-row-per-country scalar metrics (days_10_to_50, peak_deaths_pm_post50, median_age, hosp_beds_k, gdp_ppp, life_exp, and excess_mort_coverage_%) as the exclusive data source for the Page 1 hypothesis scatter visual. Finally, dash_tidy_long.csv (41,998 rows × 6 columns) melted the five core derived time-series metrics into a long key-value format (metric, value), enabling viewers to dynamically switch metrics via a single Power BI slicer while maintaining fast query performance